A PRIMITIVE P-TH ROOT OF UNITY GENERATES A GALOIS EXTENSION OF DEGREE DIVIDING P-1

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Companion to lemma-pth-root (delivered card L48357) and lemma-take-pth-root (delivered L48363): this handles the prime-to-p cyclotomic direction while those handle the purely inseparable direction.

Editorial note. The label lemma-adjoint-pth-root-unity names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter basics: a splitting field of a separable polynomial is a normal separable (Galois) extension; x^p - 1 is separable over K when p differs from char(K) (derivative p x^{p-1} is nonzero)
- Same chapter: lemma-finite-Galois (a finite Galois group has order equal to the degree); a K-automorphism of K(zeta) is determined by the image of the generator zeta

Lemma 0.1. *Let K be a field with algebraic closure $\overline{K}$. Let p be a prime different from the characteristic of K. Let $\zeta \in \overline{K}$ be a primitive pth root of 1. Then $K(\zeta)/K$ is a Galois extension of degree dividing $p - 1$.*

Proof. The polynomial $x^p - 1$ splits completely over $K(\zeta)$ as its roots are $1, \zeta, \zeta^2, \ldots, \zeta^{p-1}$. Hence $K(\zeta)/K$ is a splitting field and hence normal. The extension is separable as $x^p - 1$ is a separable polynomial. Thus the extension is Galois. Any automorphism of $K(\zeta)$ over K sends ζ to ζ^i for some $1 \leq i \leq p - 1$. Thus the Galois group is a subgroup of $(\mathbf{Z}/p\mathbf{Z})^*$. □

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